ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3, Q4

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 µm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) Assumptions: pinhole (thin-lens) camera model, lens focused on the belt plane, belt plane perpendicular to the optical axis, no lens distortion, so every pixel images the same length on the belt.

(i) The focal length in pixel units is the physical focal length divided by the pixel pitch: $f_x = f/p = (16\times10^{-3}\ \text{m})/(5\times10^{-6}\ \text{m}) = 3200$. The metres cancel so the answer is in pixels. **f_x = 3200 px**.

(ii) Width of the sensor = N·p = 1280 × 5 µm = 6400 µm = 6.4 mm. Using similar triangles between the sensor and the belt plane, $W_s/f = \text{FOV}/Z$, hence $\text{FOV} = W_s \cdot Z/f = (6.4\ \text{mm})(400\ \text{mm})/(16\ \text{mm}) = 160$ mm. **FOV = 160 mm**.

Ground sampling distance = FOV/N = 160 mm / 1280 px = 0.125 mm/px. Cross-check with the pixel-based formula Z/f_x = 400 mm / 3200 px = 0.125 mm/px; both agree, so **1 pixel = 0.125 mm = 125 µm** on the belt.

(b) Assumtions: ideal 12-bit ADC (4096 codes), the full 0–5 V range is used, no offset or gain error, and sensor noise is below one LSB so quantisation sets the limit.

Voltage step: 1 LSB = $V_{FS}/2^N = 5\ \text{V}/4096 = 1.2207\times10^{-3}$ V = 1.2207 mV.

The sensor maps force to voltage by V = S·F with S = 10 mV/N, so $\Delta F_{min} = \Delta V/S$ = 1.2207 mV / (10 mV/N) = 0.12207 N. **$\Delta F_{min} \approx 0.122$ N**. (Using 4095 steps instead of 4096 codes gives 0.1221 N, the same at 3 s.f.)

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}x(kT) + \int_0^T e^{A(T-s)}B\,u(kT)\,ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substituting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.

$x_{k+1} = 0.8187\,x_k + 0.0906\,u_k$

Sanity check: a 160 mm field of view across 1280 pixels is plausible for a 400 mm standoff and a 16 mm lens.

Question 2

Given: J = 0.02 kg·m², b = 0.05 N·m·s/rad, m = 1 kg, ℓ = 0.25 m, g = 9.81 m/s². So mgℓ = 1 × 9.81 × 0.25 = 2.4525 N·m (used throughout).

(a) Let $x_1 = \theta$ and $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$; $\dot{x}_2 = (1/J)[\tau + b\,x_2 - mg\ell \sin x_1]$.

Equilibrium: $x_2 = 0$ and $mg\ell \sin\theta_e = \tau_0$. Because $-1 \le \sin\theta \le 1$ we need $|\tau_0| \le mg\ell = 2.4525$ N·m.

(b) Jacobian at $\theta_e = 60°$: $A = [0,\ 1;\ -(mg\ell/J)\cos\theta_e,\ -b/J]$.

$mg\ell/J = 2.4525/0.02 = 122.625$; $\cos(60) = -0.9524$, so $A_{21} = -122.625 \times (-0.9524) = +116.8$. $A_{22} = -b/J = -2.5$.

A = [0, 1; 116.8, −2.5], **B = [0; 50]**.

(c) Set all time derivatives to zero in the closed loop. With $\tau = K_p(\theta_d - \theta) - K_d\dot{\theta}$ and $\dot{\theta} = \ddot{\theta} = 0$ the damper term vanishes (so K_d has no effect on the final position) and the dynamics reduce to $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss})$.

Why $\theta_{ss} \neq \theta_d$: if the joint were at rest exactly at $\theta_d = 90°$ the PD torque would be $K_p \cdot 0 - K_d \cdot 0 = 0$, yet graviy pulls with $mg\ell \sin 90° = 2.4525$ N·m. The net torque would be non-zero, so the joint could not be at rest. A purely proportional-derivative law can only hold a load if there is a stedy error.

Let $e = \theta_d - \theta_{ss}$ (in rad). Then $\sin\theta_{ss} = \sin(\pi/2 - e) = \cos e$, and the equation becomes $50\, e = 2.4525 \cos e$. Fixed-point iteration: $e_0 = 2.4525/50 = 0.04905$ (taking $\cos e \approx 1$); $e_1 = 0.04905 \times \cos(0.04905) = 0.04899$; $e_2 = 0.04905 \times \cos(0.04899) = 0.04899$, converged.

$e = 0.04899$ rad $\times (180/\pi) = 2.807°$, so $\theta_{ss} = 87.19°$. **Steady-state error ≈ 2.81° (the arm sags below the target).** This is small because K_p is much larger than $mg\ell$ ($K_p/mg\ell \approx 20$).

(d) Stedy state: $\dot{\theta} = \ddot{\theta} = 0$ and the control law gives $\tau = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$. Plugging into the plant, $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$, so $K_p(\theta_d - \theta_{ss}) = 0$ and **$\theta_{ss} = \theta_d$: steady-state error is zero**.

Poles: linearise the closed loop about θ_d. Before compensation, $J\ddot{\theta} + (b + K_d)\dot{\theta} + (K_p + mg\ell \cos\theta_d)\,\delta\theta = 0$, characteristic polynomial $J\lambda^2 + (b + K_d)\lambda + K_p + mg\ell \cos\theta_d$. After compensation the added term $mg\ell \sin\theta$ cancels the plant's graviy term exactly, leaving $J\lambda^2 + (b + K_d)\lambda + K_p$.

The two differ only by $mg\ell \cos\theta_d$, and for $\theta_d = 90°$ this is $2.4525 \times 0 = 0$. So **the closed-loop pols are unchanged** at this operating angle. I should stress this is becuase $\cos 90° = 0$; if θ_d were, say, 60°, the compensation would raise the stiffness by 61.3×0.02... i.e. it would move the poles.

Question 3

Given: $A = [0, 1; 0, -2]$, $B = [0; 1]$, $C = [1, 0]$, $f_s = 1$ kHz so the sample time is $T = 1/f_s = 1$ ms.

(a) Reason 1 – noice amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequncy ($\approx \omega$ for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly $1/T = 1000$ s^{-1}, and can swamp the true velocity.

Reason 2 – quantisaton: the encoder only reports integer counts, so $\Delta\theta$ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) $\omega_c = 20$ rad/s. Tustin with $T = 0.001$: $a = \omega_c T/2 = 0.01$.

$y_k = [a/(1 + a)](u_k + u_{k-1}) + [(1 - a)/(1 + a)]\, y_{k-1}$ = **$0.00990\,(u_k + u_{k-1}) + 0.98020\, y_{k-1}$**.
DC gain: $2(0.0099)/(1 - 0.9802) = 0.0198/0.0198 = 1$ ✓.

(c) First check observability: $O = [C; CA] = [1, 0; 0, 1]$, full rank, so the observer poles can be placed arbitrarily.

With $L = [\ell_1, \ell_2]^T$ and $C = [1, 0]$, $LC = [\ell_1, 0; \ell_2, 0]$, so $A - LC = [-\ell_1, 1; -\ell_2, -2]$.

Characteristic polynomial: $\det(sI - (A - LC)) = \det[s + \ell_1, -1; \ell_2, s + 2] = (s + \ell_1)(s + 2) - (-1)(\ell_2) = s^2 + (2 + \ell_1)\, s + (2\ell_1 + \ell_2)$.

Both estimator poles at $s = -10$ means the desiered polynomial is $(s + 10)^2 = s^2 + 20\, s + 100$.
Comparing coefficients: s^1: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$. s^0: $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 36 = 64$.
$L = [18, 64]^T$. Verification: $A - LC = [-18, 1; -64, -2]$ has trace -20 and determinant $(-18)(-2) - (1)(-64) = 36 + 64 = 100$, so the polynomal is $s^2 + 20s + 100 = (s + 10)^2$ ✓.

(d) The observer is more accurate than the filter.

The filter is safer when you need something simple and cheap to implement.

Question 4

Given: plant $\theta\ddot{} + a\theta\dot{} + c\theta = b\,u$ with $a = 2$, $c = 6$, $b = 1$; controler $u = N_{ff}\,r - k_1\theta - k_2\theta\dot{}$.

(a) Closed loop: $\theta\ddot{} + (a + bk_2)\theta\dot{} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $s^2 + 4s + 8 = 0$ so the pols are $s = -2 \pm 2j$.

DC gain $= bN_{ff}/(c + bk_1) = N_{ff}/8$. For unity gain $N_{ff} = 1/8 =$ **0.125**.

(b) With the actual $b = 0.75$ and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\theta\ddot{} + (a + bk_2)\theta\dot{} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\theta\ddot{} + 3.5\theta\dot{} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 =$ **0.8**.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 =$ **20%**. A common mistake would be to say the output falls by the same 25% as the acuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, wich decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stable.

(c) Substituting the control law and differentiating once: $s^3 + (c + bk_2)s^2 + (a + bk_1)s + bK_I$.

Desired: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$6 + k_2 = 8 \Rightarrow \mathbf{k_2 = 2}$; $2 + k_1 = 24 \Rightarrow \mathbf{k_1 = 22}$; $K_I =$ **32**.

(d) Suppose the closed loop is stable and r is constant. Then the state converges to a constant, so $\theta\dot{} = \theta\ddot{} = 0$ and, crucially, $\xi\dot{} = 0$ as well (ξ itself is constant at its final value ξ_{ss}). From the integrator equation $\xi\dot{} = r - \theta$, we get $r - \theta_{ss} = 0$, i.e. $\boldsymbol{\theta_{ss} = r}$.

Neither b nor c appears in this argument. They only influence the value of ξ_{ss} needed to hold the equilibrium (from the plant equation $c\theta_{ss} = bu_{ss} = b(-k_1\theta_{ss} + K_I\xi_{ss})$), and the pole locations, so the zero error survives any error in the model parameters provided the loop stays stable (unlike the feed-forward N_{ff} scheme of part (b)).

Practical failure mode: integrator windup. If the actuator saturates (torque limit), the loop is effectively open, the error stays non-zero and ξ keeps integrating to a huge value. When the reference becomes reachable the stored integral forces a large overshoot and a long recovery. Standard remidy: anti-windup, such as clamping the integrator when the acuator is saturated, conditional integration, or back-calculation that feeds the difference between commanded and saturated torque back to the integrator.